

Day 1

Effective population size (N_e)



Disclaimer – What to expect



Few formulas



In-depth explanations
or brief overviews



Applied examples

Outline

1

Genetic / genomic data

Genetic variability

Genetic drift

Effective population size (N_e)

Approaches to infer N_e from genomic data

2

Methodology 1:
coalescent-based approach

Tutorial – part 1

Methodology 2:
LD-based approach

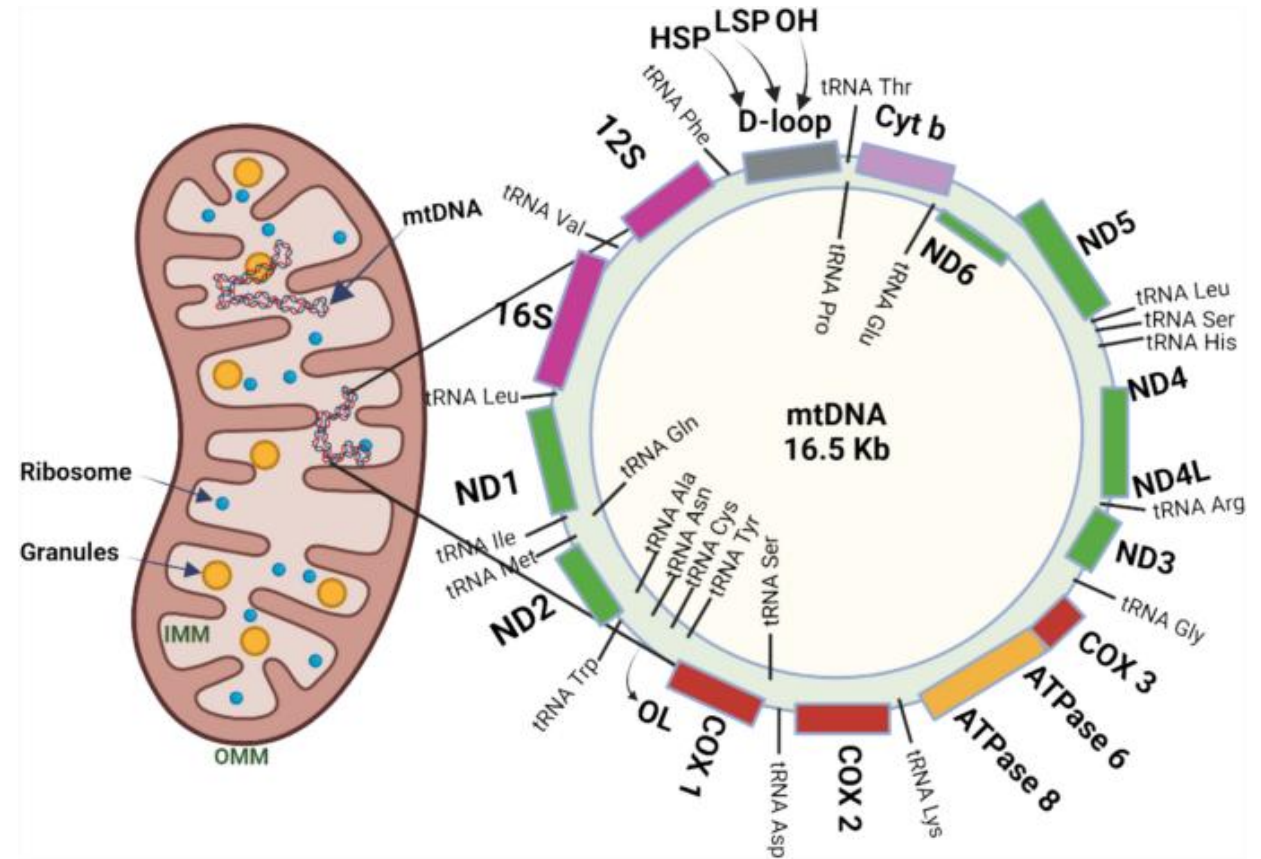
Tutorial – part 2

3

Discussion
(with Dr. Roberta Gargiulo)

Genetic / genomic data types

- Mitochondrial DNA (mtDNA)



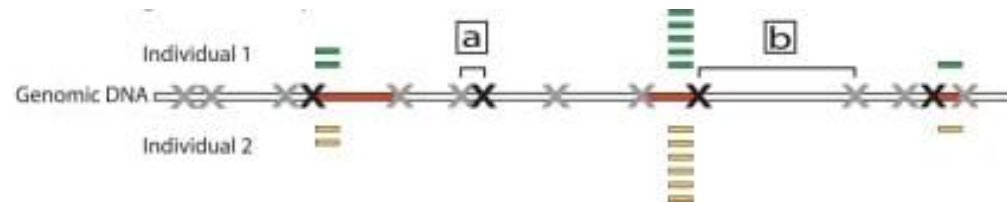
Genetic / genomic data types

- Mitochondrial DNA (mtDNA)
- Microsatellite loci (or short tandem repeats, STR)



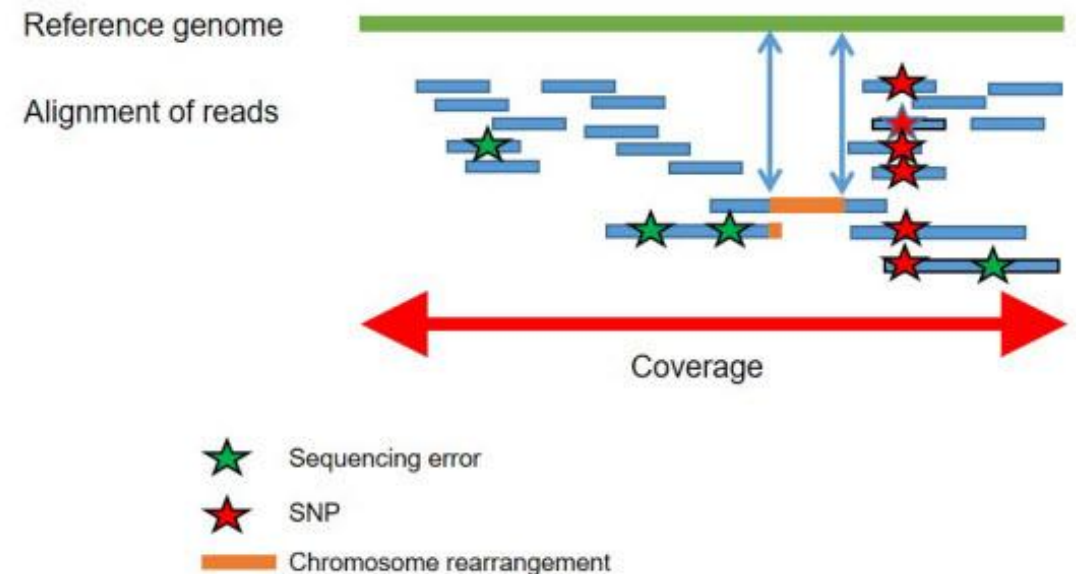
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- Mitochondrial DNA (mtDNA)
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- (double digest) Restriction-site associated DNA sequencing (ddRAD)



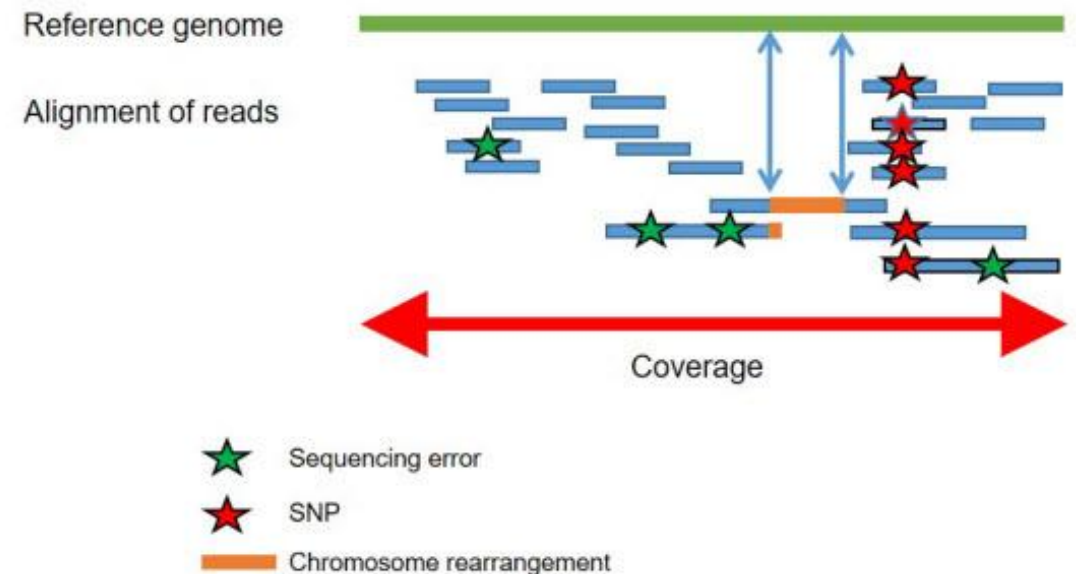
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Genetic / genomic data types

- Mitochondrial DNA (mtDNA)
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- Whole genome resequencing (WGS)
- SNPchip
- RNA sequencing (RNAseq)
- Environmental DNA
- Ancient DNA



Genetic variability

01

What is it?

02

**Why is it
important?**

03

**How do we
measure it?**

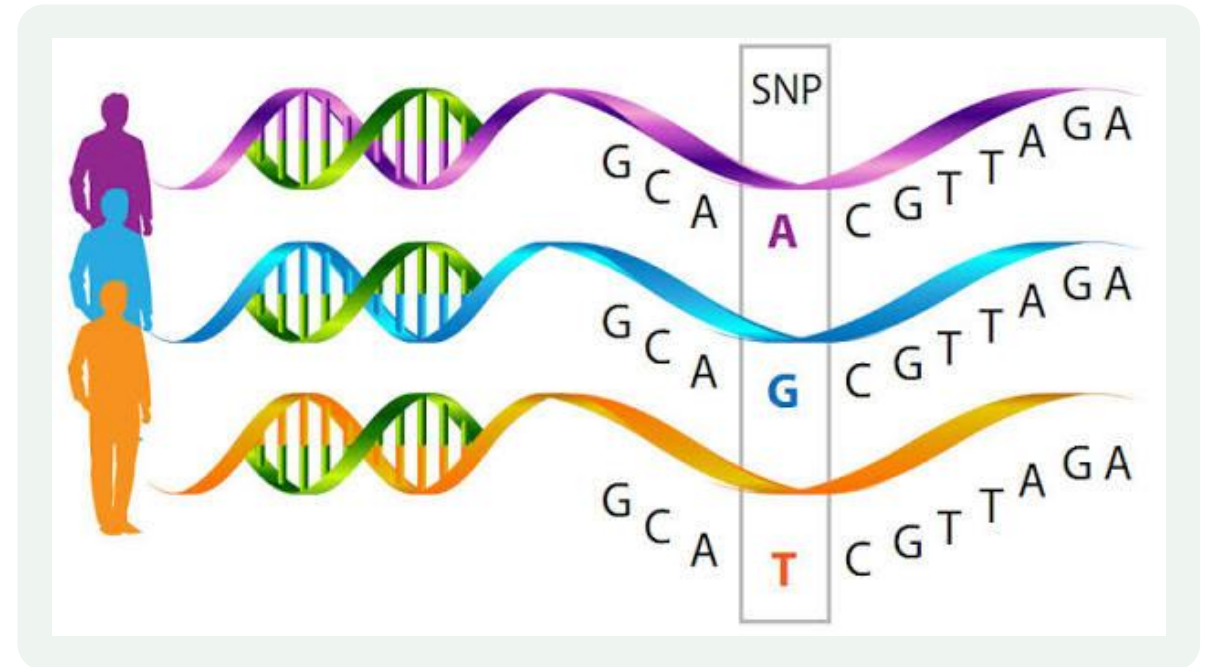
04

**How does it
vary through
time?**

Genetic variability – What is it?

Genetic variability (or genetic diversity):

differences along DNA sequences resulting in phenotype variability among individuals and populations.



Genetic variability – Why is it important?



**Fuel for Adaptive
Evolution**



**Prevention of
Inbreeding Depression**



**Conservation and
Extinction Risk**



**Selective Breeding &
Domestication**

Genetic variability – How do we measure it?

- Expected heterozygosity (H_e)
- Nucleotide diversity (π)
- Watterson's theta (θ_w)
- Allelic richness (R)
- Runs of homozygosity (ROH)

Sample 1	A	T	A	G	C	A	A
Sample 2	A	T	T	G	G	A	A
Sample 3	A	T	A	G	C	A	A
Sample 4	A	T	A	G	C	A	T
Sample 5	A	T	A	G	G	A	T

Genetic variability – How does it vary through time?

Core Evolutionary Forces



Mutation

The continuous, ultimate source of all new genetic variation.



Genetic Drift

The random loss of alleles and reduction of heterozygosity over time, occurring at a rate inversely proportional to the effective population size ($1/N_e$)



Natural Selection

Directional selection locally reduces diversity around selected sites via genetic hitchhiking, whereas balancing or frequency-dependent selection maintains variation over time



Gene Flow & Admixture

Restores or introduces novel alleles between populations, counteracting the diversity-depleting effects of drift

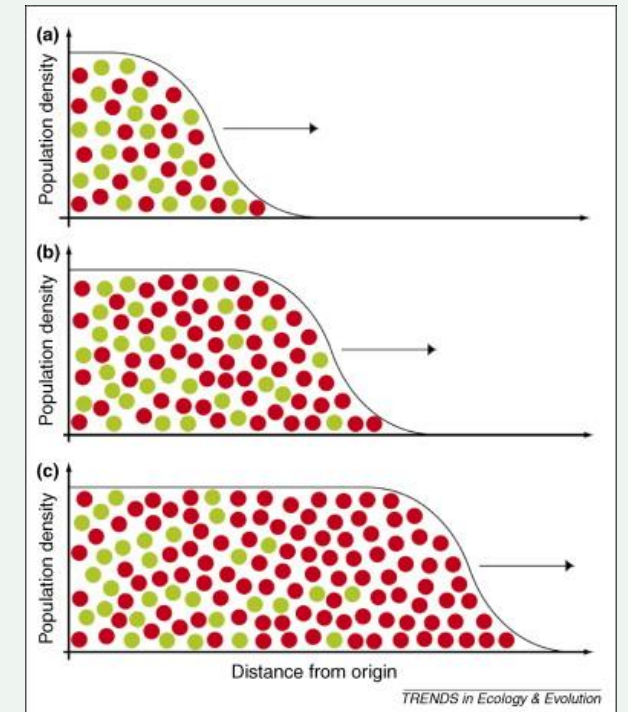
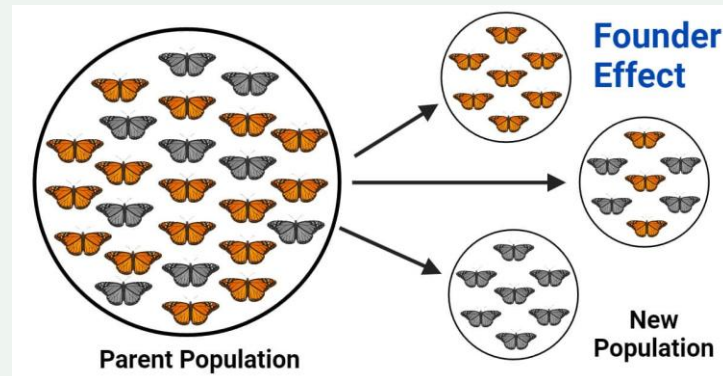
Genetic variability – How does it vary through time?

Demographic changes

- Population bottleneck
- Founder effect
- Range expansion

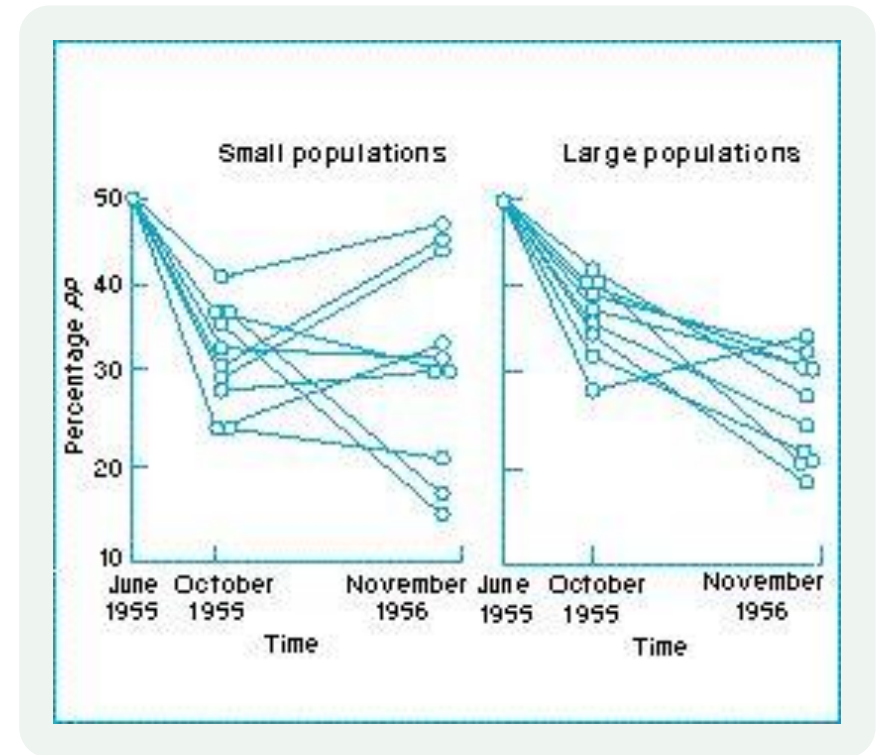


Increased genetic drift



Genetic drift – What is it?

- **Definition:** It is the random change in allele frequencies from one generation to another caused by «sampling error» during gamete transmission in finite populations
- **Consequence:** It leads to the random loss of genetic variation (heterozygosity) and an increase in inbreeding or allele fixation over time
- **Property:** It is stronger on smaller populations



$$\text{Var}(\Delta q) = \frac{q(1 - q)}{2N_e}$$

What is N_e ?

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Effective Population Size



Medium ground finch, Example 7.2

Effective population size is whatever must be substituted in the formula $(1/2N)$ to describe the actual loss in heterozygosity.

Sewall Wright (1969)

Effective population size (N_e) is one of the most fundamental evolutionary parameters of biological systems, and it affects many processes that are relevant to biological conservation.

Robin Waples (2002)

What is N_e ?



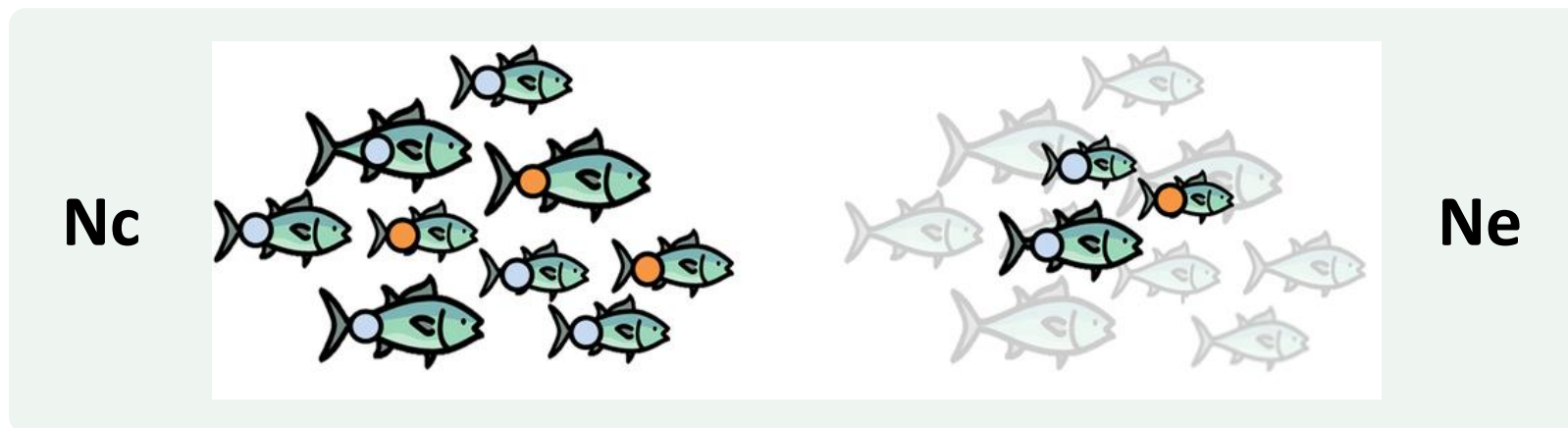
«**Effective population size (N_e)** is defined as the size of the ideal population (N) that will result in the same amount of genetic drift as in the actual population being considered.»

Wright-Fisher idealized population

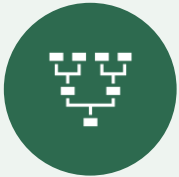
- ✓ large population with constant N
- ✓ non-overlapping generations
- ✓ diploid monoecious individuals
- ✓ panmictic, random mating
- ✓ no mutation, selection, migration

Relationship between N_e and N_c

- **Census population size** (N_c or N) is the actual count of individuals in a population, which governs ecological processes such as competition, predation, and population growth
- In natural populations, N_c is almost always larger than N_e
- N_e/N_c ratio in wild species typically falls between 0.1 and 0.2.



Main Drivers Reducing N_e relative to N_c



Variance in Reproductive Success

High variance in family size among parents increases the rate of genetic drift, substantially reducing N_e .



Skewed Sex Ratios

Unequal numbers of breeding males and females restrict N_e toward four times the size of the rarer sex.



Fluctuations in Population Size

Across generations, long-term N_e is calculated as the harmonic mean of single-generation sizes, so past population bottlenecks disproportionately depress N_e .



Demographic Complexities

Overlapping generations, spatial structure, and non-random mating

Why is N_e important in conservation?



Quantifying the Loss of Genetic Diversity

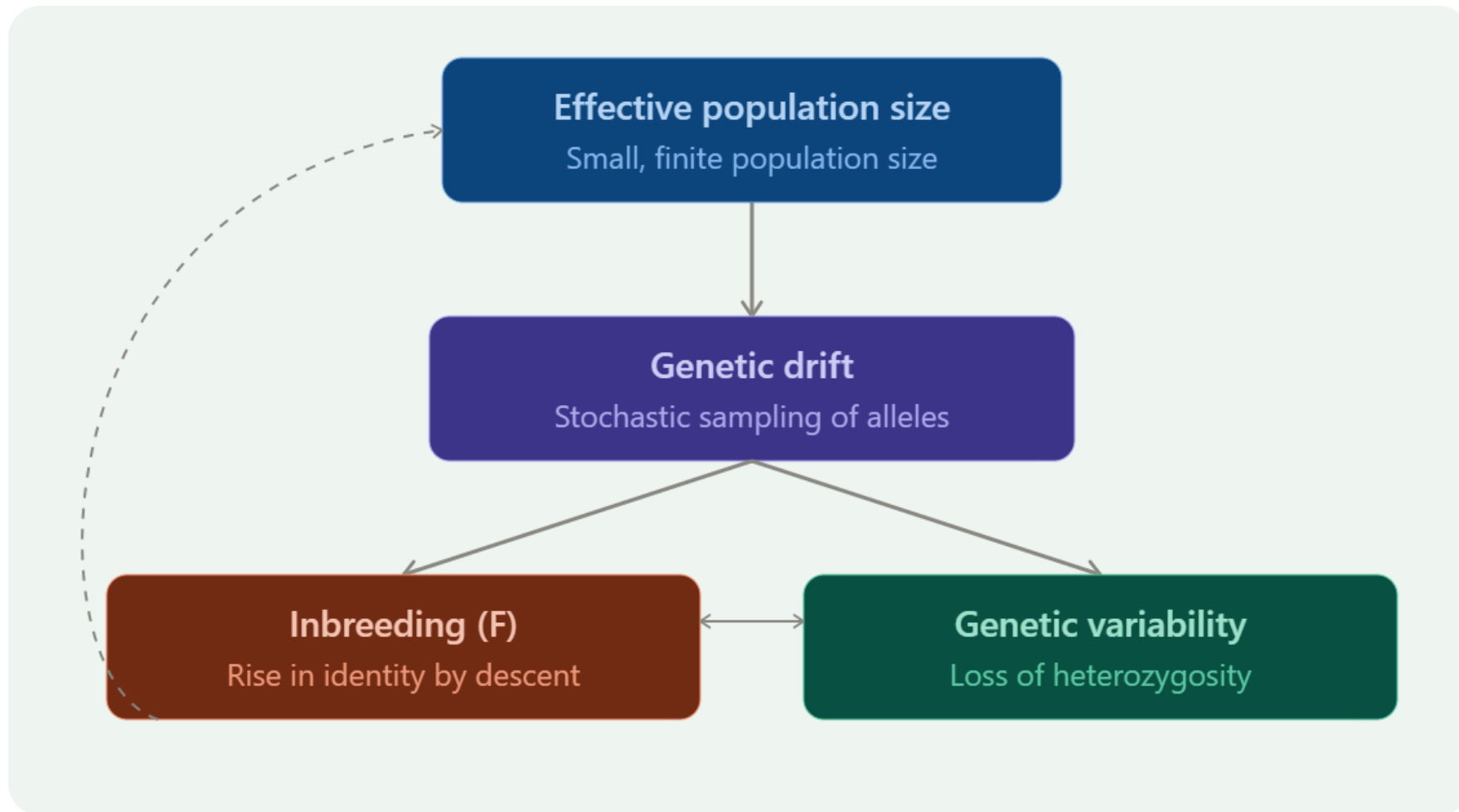
Genetic drift causes the random loss of alleles and reduces heterozygosity at a rate inversely proportional to the effective population



Genetic Rescue

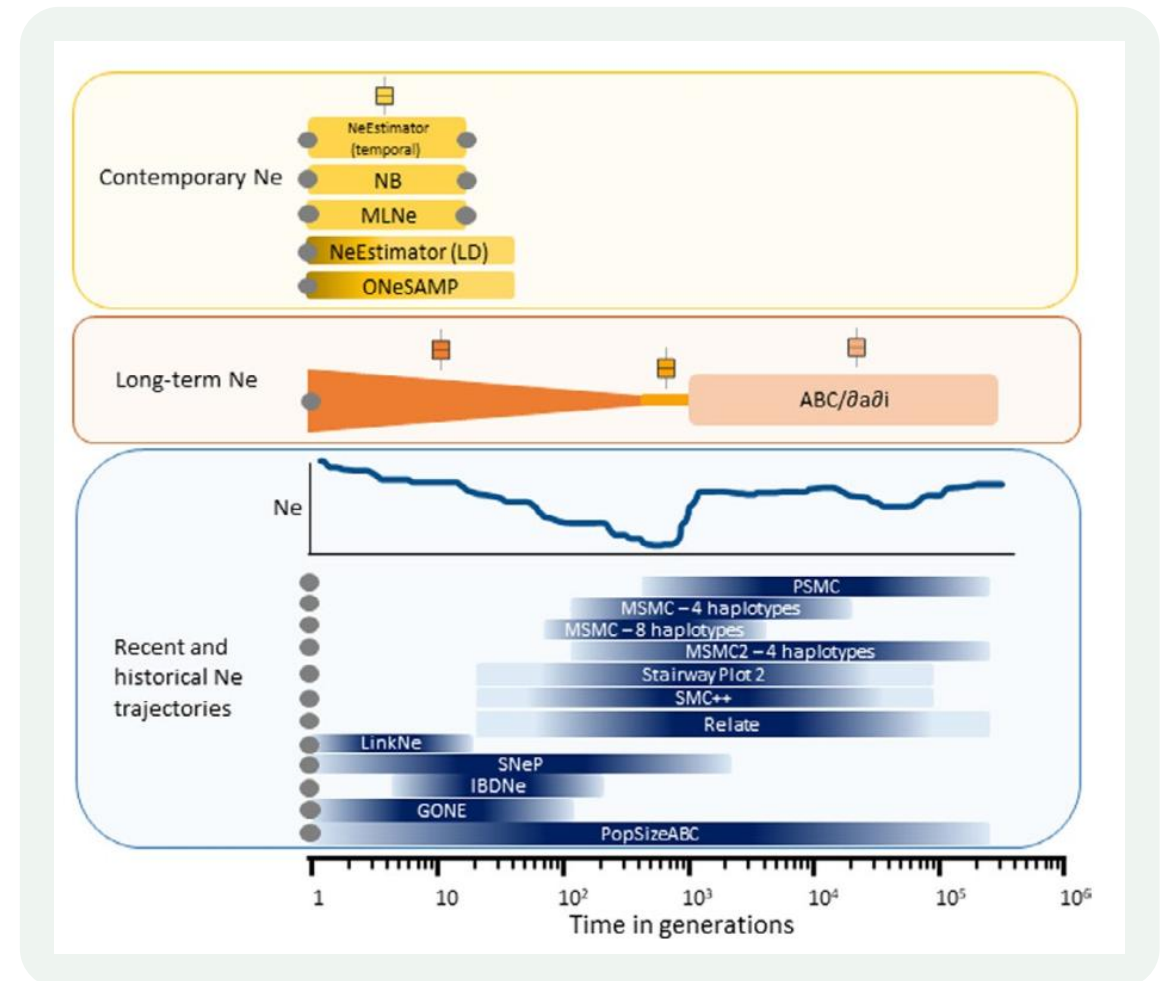
Estimating N_e helps identify isolated, highly inbred populations that require an intervention through controlled gene flow to sustain genetic diversity

Relationship between processes and measures



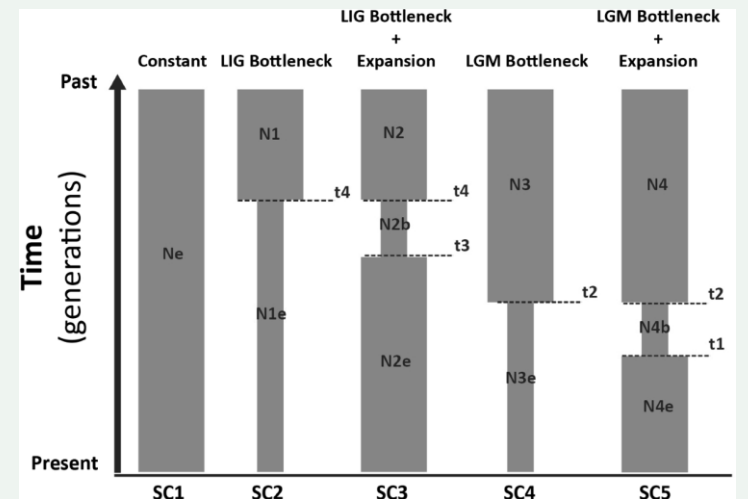
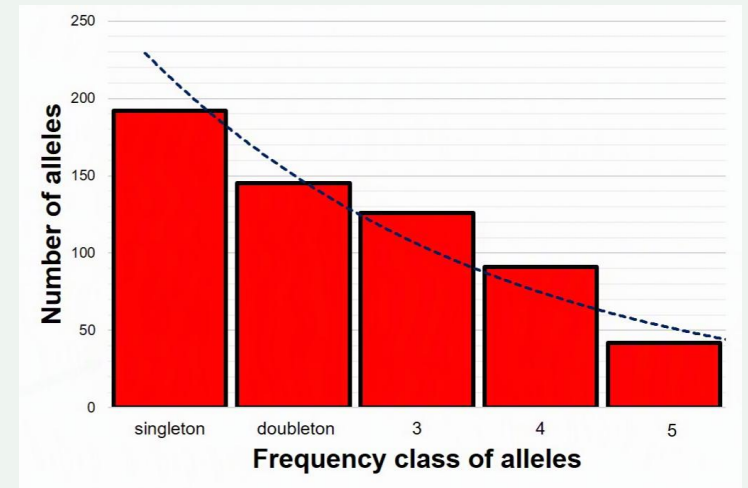
Main approaches to estimate Ne

- Coalescent
- Linkage Disequilibrium (LD)
- Identity By Descent (IBD)
- Site Frequency Spectrum (SFS)
- Approximate Bayesian Computation (ABC)



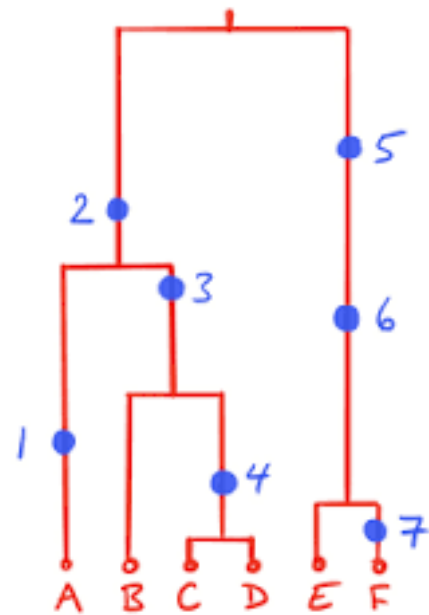
How do we measure historical N_e ?

- **Site Frequency Spectrum (SFS)**: different demographic histories create different patterns of genetic variation
- **Approximate Bayesian Computation (ABC)**: summary statistics are computed on the real data set and from an ensemble of coalescent simulations to evaluate which demographic scenario has the highest posterior probability of the generating the observed summary statistics
- **Coalescent**: distribution of heterozygous sites across the genome contains information about the set of genealogies that gave rise to the genome

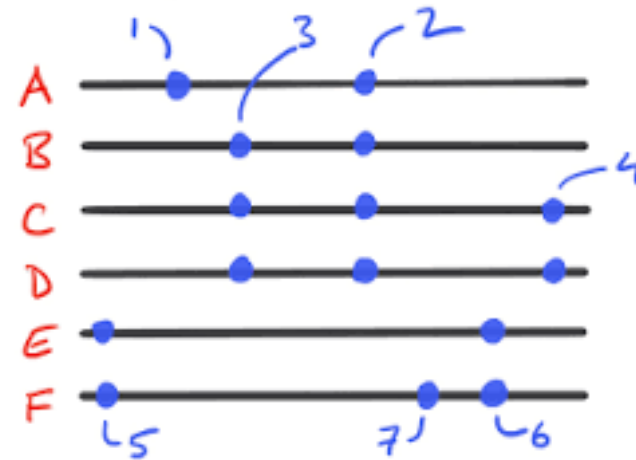


Coalescent-based methods

A. TREE WITH MUTATIONS

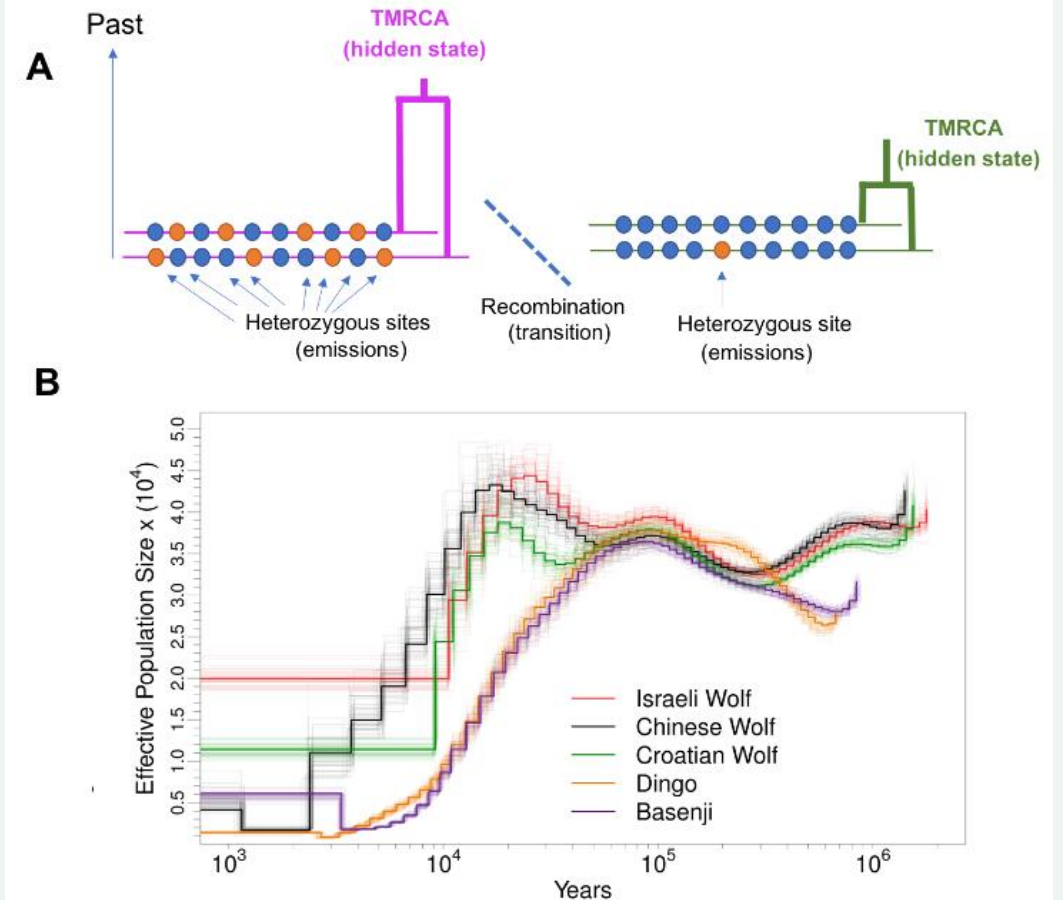


B. HAPLOTYPES



Coalescent-based methods

- PSMC uses the presence of SNPs across genomic fragments to infer the distribution of coalescent times using a hidden Markov model
- MSMC allows to investigate more recent past because more haplotypes mean more recent coalescent events possibly
- scm++ increases even more the possibility of describing recent times with the inclusion of the SFS from many samples
- Pros: only one (PSMC) or few (MSMC) genomes
- Cons: high coverage, prone to error in case of gene flow



Data quality and filters



High coverage (at least 18x)

Lower coverage might miss heterozygous calls biasing the inference



Mask genes and repetitive regions

Genes may be subjected to non-neutral evolution, repetitive regions may inflate heterozygosity

Hard filtering

From the raw VCF with variant AND invariant sites, apply filters to keep only high-quality positions:

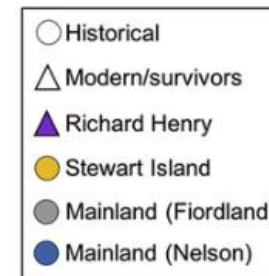
- ✓ Exclude indels and e.g. ± 5 bp around them
- ✓ Min and max depth (± 2 or 3 from average depth)
- ✓ Missingness (max 10% per site and sample)
- ✓ Mapping, genotype, and base quality (between 30 and 60)
- ✓ For other GATK filters check [here](#)



Example 1: ancient demographic reconstruction of the kākāpō

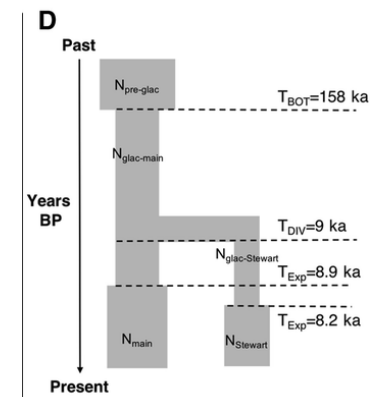
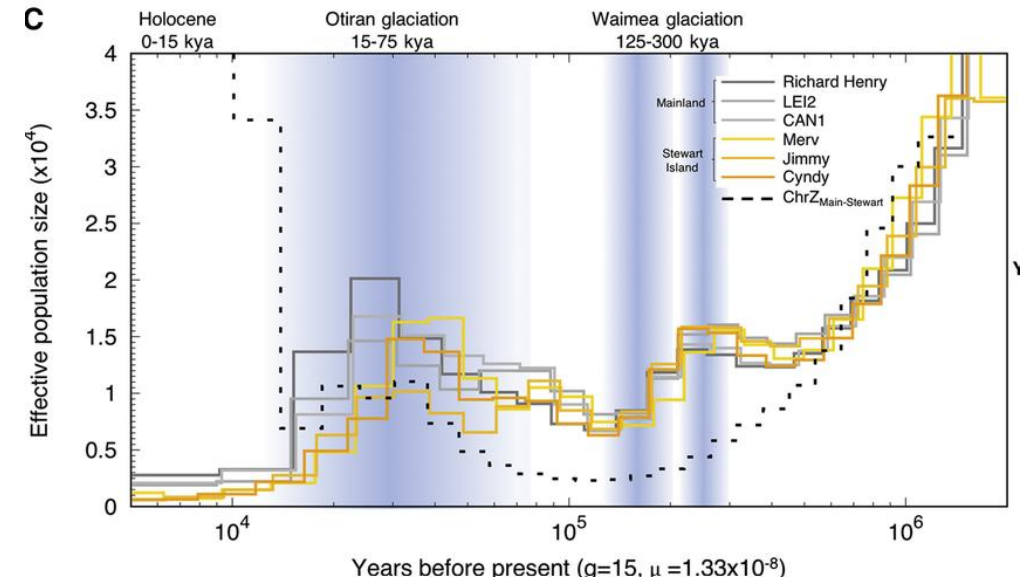
- New Zealand colonizations: 1360 CE and 1800s
- Overhunting and introduction of mammalian predators
- By 1995, the species was reduced to 51 birds
- In 2021, a total of 201 kākāpō in island sanctuaries

A



Example 1: ancient demographic reconstruction of the kākāpō

- New Zealand colonizations: 1360 CE and 1800s
- Overhunting and introduction of mammalian predators
- By 1995, the species was reduced to 51 birds
- In 2021, a total of 201 kākāpō in island sanctuaries
- Drop in population well before human settlements
- Stewart island was colonized much earlier than previously thought



How do we measure recent N_e ?



Temporal method

sample populations at two time points (distant t) and compare their allele frequencies



Sibship method

a population with a small N_e also produces a large proportion of close relatives (sibs)



Identical By Descent (IBD) tract length method

individuals from small populations are expected to share longer chromosome tracts that are IBD



Linkage disequilibrium method (LD)

genetic drift creates association of alleles across loci with an inverse relationship with N_e

LD-based methods

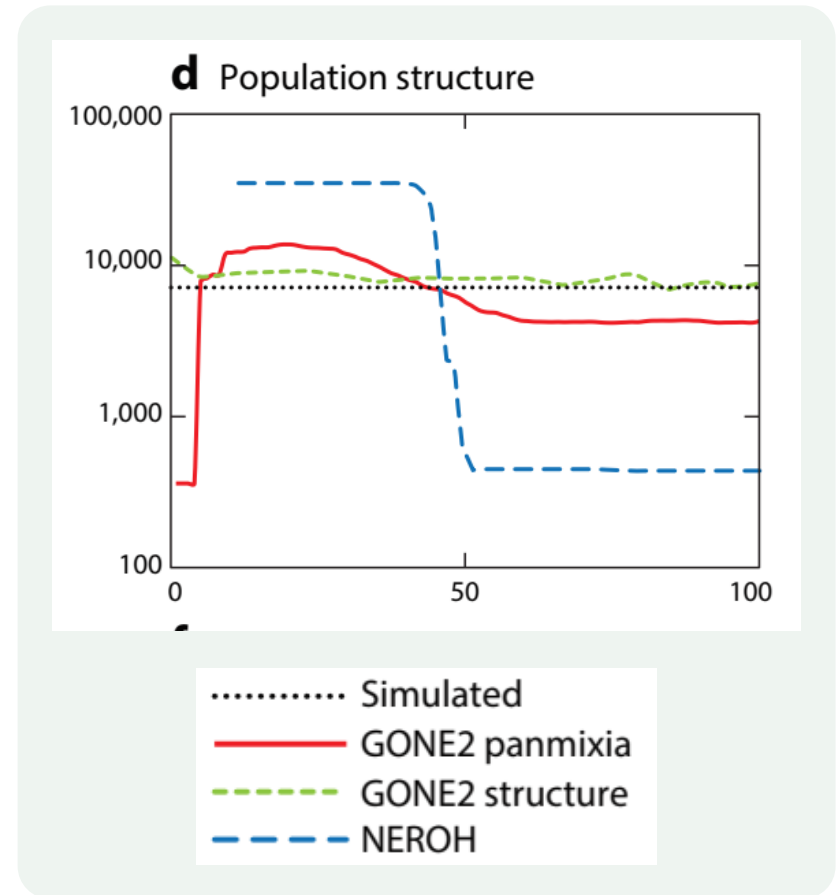
- LD refers to the nonrandom association of alleles at two loci
 - An allele at one locus may be inherited with a particular allele at another locus more often than expected
 - Recombination can break the association, but it is harder if the loci are tightly linked
 - LD methods leverage the following relationship: $r^2 \approx \frac{1}{4N_e c + 1}$
-
- Pros: time-point samples, no phased haplotypes
 - Cons: recombination map, large sample size, population structure, quality of sequencing data

GONE2: historically recent N_e estimation

- LD between closely linked markers is affected primarily by ancient demographic changes
- LD between loosely linked markers is influenced by more recent demography

→ Inverse relationship between genetic distance and coalescent time: $c \approx 1/(2t)$

- Note on the timescale: How far back in time?
It depends on the resolution of the genetic map, usually reliable estimates for up to 100–200 past generations



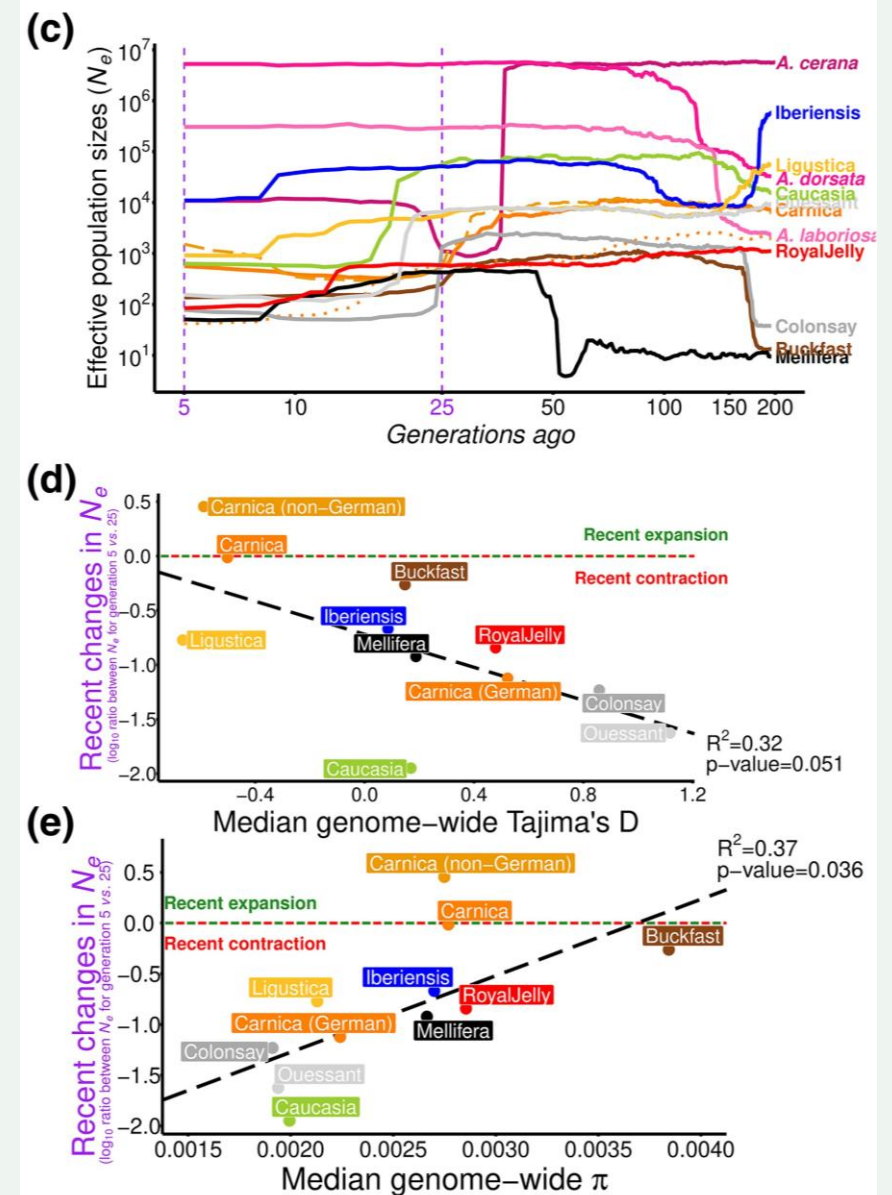
CurrentNe2: contemporary Ne estimation

- It accommodates high population sizes and low sample sizes
- It resolves the lack of recombination map by providing an estimate integrated over the entire genome, assuming the SNPs are randomly scattered across chromosomes that are each 1 Morgan long

Methods (software)	Cattle ^j
Temporal method (NeEstimator V2) ^b	NA
Temporal method (MLNe) ^c	NA
LD panmixia (CurrentNe2) ^d	12 (12–12)
LD structure (CurrentNe2) ^e	1,770 (1,748–1,792)
Sibship method (COLONY) ^f	68 (49–95)
Heterozygosity excess (COLONY) ^f	∞ (0– ∞)
Molecular coancestry (NeEstimator V2) ^g	NA
ABC method (ONeSAMP 3.0) ^h	25.2 (24.8–26.1)

Example 2: *Apis mellifera* management in Europe

- *Apis mellifera* (692 samples) and 3 other *Apis* species (54 samples)
- 3 *A. mellifera* lineages, one particularly preferred for production
- Sanctuaries are reducing genetic diversity after the bottleneck for their creation
- Heavy selection for traits creates differences in 20 generations



Recommended readings and resources



- Allendorf, F. W., Luikart, G. H., & Aitken, S. N. (2012). *Conservation and the genetics of populations*. John Wiley & Sons.
- Caballero, A., Santiago, E., & Wang, J. (2026). Estimation of Contemporary and Recent Effective Population Size from Genomic Data. *Annual Review of Ecology, Evolution, and Systematics*, 57.
- Nadachowska-Brzyska K, Konczal M, Babik W. 2022. Navigating the temporal continuum of effective population size. *Methods Ecol. Evol.* 13:22–41
- Waples, R. S. (2025). The idiot's guide to effective population size. *Molecular Ecology*, 34(23), e17670.
- <https://ccgenetics.github.io/guidelines-genetic-diversity-indicators/>
- <https://github.com/stschiff/msmc-tools/tree/master>